

Lecture Notes on QFT I

November 13, 2025

Readings

- Peskin & Schroeder Sec. 2.1 and 2.2
- Weinberg vol 1 Chap. 1

Lecture 1: Principle of Locality and Fields

The goals of this Quantum Field Theory class is

- To develop concepts, formalism and techniques for quantum dynamics of fields.
- Quantum Field Theory is also very important for other interactions in nature. Among the four fundamental interactions in nature, three of them are described by QFT completely.
- Even though QFT is initially developed for particle physics, but over the years, QFT has been found many applications in many other branches of physics, in condensed matter, statistical physics and so on. It's fair to say, nowadays, QFT has become a universal language for theoretical physics in all different fields.

Chapter 1. Why we consider QFT?

principle of locality

The first important concept is called the **principle of locality**.

In Newtonian mechanics, you have action at a distance, for example, if you look at gravity and the gravity is exerted by the sun on the Earth, but they are very far away. And the same thing with the Coulomb interaction between the charged particles.

But then, in the 19th century, they came from this principle of locality, that is formulated by Faraday around 1830. The principle of locality says that all points you don't have action at a distance. He said **all points in space participate in the physical process, and the effect propagates from points to a neighboring points**. So in this principle of locality, you don't have action at a distance, action at a distance is always conveyed from one point to another point through the propagating in the space.

The concepts of **fields** is **the mathematical device or vehicle that the principle locality is at work**. The main idea of the field is that we associate each point in space a dynamical variable.

For example, if you have electric field is defined at each point x , we can introduce an electric field, which can also depend on time. Of course, normally we write it this way $\vec{E}_{\vec{x}}(t) \equiv \vec{E}(\vec{x}, t)$. The reason I write it in this way $\vec{E}_{\vec{x}}(t)$ to emphasize that in the definition of the electric field, the space and times actually play very different roles. The space plays the role of a label. So at each point $\vec{x}$, we have electric field, so x here is just a label, and t is used to describe the evolution, the change of the electric field. So $\vec{x}$ and t play very different roles.

Similarly, you can do this to magnetic field $\vec{B}_{\vec{x}}(t) \equiv \vec{B}(\vec{x}, t)$, you should always treat $\vec{x}$ as labels. And we know the evolution of electric and magnetic field is described by Maxwell's equations,

$$\begin{cases} \nabla \cdot \vec{B} = 0 \\ \nabla \cdot \vec{E} = 4\pi\rho \\ \frac{\partial \vec{B}}{\partial t} = -\nabla \times \vec{E} \\ \nabla \times \vec{B} = 4\pi\vec{J} + \frac{\partial \vec{E}}{\partial t} \end{cases} \quad (1)$$

which is called the differential form of the Maxwell's equations. This set of equations exemplifies perfectly the principle of locality. It's because you see those equations only involve the value of electric fields and the magnetic field at a single point. If $\vec{B}$ is at $\vec{x}$ point, then $\vec{E}$ is also at $\vec{x}$ point. You never say $\vec{B}_{\vec{x}}$ but $\vec{E}_{\vec{y}}$. So this

reflects the principle of locality that the effect is propagating from point to point, you just have the derivative of the same point. You don't involve separate points. So we call Maxwell's equations as local equations, they contain only E and B at same point with finite number of derivatives. So the derivatives are the ones to help you to propagate because it relates the point to the neighboring point.

Another example of locality is the Einstein's general relativity. In Einstein gravity, the dynamical variables are so-called spacetime metric $g_{\mu\nu}(\vec{x}, t)$. They are object with two indices, two spacetime indices, and the Einstein's equations are equations for this kind of object $g_{\mu\nu}$. And again this equation are local equations, in the sense that they only depends on the g evaluated at the same point with a finite number of derivatives. So in Einstein's gravity, you no longer have action at distance, so the effect of the gravity is propagated through the spacetime.

Here we say these two examples exemplify the principle of locality. In fact, the principle of locality plays a very important role in formulating those equations, which is a very powerful principle.

Lecture 2: Classical Field Theories

Types of field

- A scalar field $\phi(\vec{x}, t)$. This is a quantity defined at each point; there is only one value at the given point. For example, the temperature is a single scalar quantity defined at each point.
- A vector field: $\vec{E}(t, \vec{x})$ and $\vec{B}(t, \vec{x})$ are examples of vector fields because you have a vector at each point. Here, I'm using the three-dimensional notations, but in the relativistic theory, we can use the four vector notations, e.x., the vector potential has the form $A_\mu(x^\nu)$. Here A_μ is a four-vector, obtained by combining the spatial and temporal components into a single spacetime vector.
- A tensor field: $g_{\mu\nu}(x^\lambda)$ is an example for tensor field. Here you have two indices, you have many different components depend on the dimension. At each point, you have an object with two indices.
- Spinor fields $\psi_\alpha(t, \vec{x})$, where α is another index that we will define later.

Convention

- μ is always from 0 to 3, and sometimes I write simply x ; when I write $x = x^\mu = (ct, \vec{x}) = (ct, x^i)$, it denotes a four-vector. The spatial index is always denoted by i .
- $\partial_\mu = (\frac{1}{c}\partial_t, \nabla)$, ∇ is the gradient on the spatial directions.
- A four-vector $A_\mu = (A_0, \vec{A}) = (A_0, A_i)$, we still use i to denote the spatial components.
- $c, \hbar = 1$.

Action Principle in Classical Mechanics

First, recall in your classical mechanics, we introduce the action

$$S = \int dt L(x, \dot{x}, t), \quad (2)$$

where Lagrangian is a function of x , $\dot{x}$ and the time. This is the one-dimensional particle motion. And you can also introduce the canonical momentum, which is defined by

$$p = \frac{\partial L}{\partial \dot{x}}. \quad (3)$$

The Hamiltonian is related to the Lagrangian by

$$H = p\dot{x} - L, \quad (4)$$

through a Legendre transformation. And the equation of motion is obtained by extremizing $S[x(t)]$, where S can be consider as a functional of your trajectory, so whatever trajectory you have, you extremize this S , and you obtain the equation of motion (EOM).

Classical Field Theory

Now we can generalize this to field theory. From the principle of locality, the form of the Lagrangian L in field theory is significantly constrained. It must have the form:

$$L = \int d^3x \mathcal{L}(\phi_a, \partial_t \phi_a, \partial_i \phi_a) \quad (5)$$

which is the integration over all spatial directions. Here, $\phi_a(\vec{x}, t)$ is a function of spatial direction and time, it is a general field, and a label different fields. e.g., a can label different scalar fields if you have different scalar fields, a can also refer to A_μ or to spinor fields, and so on. $\mathcal{L}$ is called the Lagrangian density, which depends only on the value of ϕ_a and its derivatives at a single point. They are still local.

Remarks: Why the Lagrangian must have this form?

- First, the principle of locality implies (5) only involves a single integral because the locality does not allow something like this,

$$L = \cdots + \int d^3x d^3y \phi(\vec{x}, t) \phi(\vec{y}, t) \quad (6)$$

which involving the ϕ at different points. If your Lagrangian contains such terms, then when we derive the equations of motion, they will not be local. EOM will involve the behavior of your field at one point and then influenced by the field value at a distant point, then you will not have local equation.

- We only allow first derivative in time, we don't allow high order derivatives in time. The reason is that again the equation of motion will involve more than two derivatives in time if your Lagrangian has high order derivatives in time. $\partial_t \phi$ in (5) implies that EOMs contain two t -derivatives, which come from our experiences. In real life, all experiments are determined by the initial conditions. For the initial condition, you only need to specify the position and velocity; no higher derivatives are required. If the equation of motion involves more than two derivatives in time, then you need to specify more general initial conditions. But for simplicity, for the most of the time, as we will see, we will restrict to quantum field theory in first derivative in time.

So now as in classical mechanics, we can introduce the canonical momentum, Hamiltonian, and so on. Here, we can introduce the so-called canonical momentum density

$$\Pi_a(\vec{x}, t) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}_a(\vec{x}, t)}, \quad \text{where} \quad \dot{\phi}_a(\vec{x}, t) = \partial_t \phi_a \quad (7)$$

So this is just the direct generalization of classical mechanics. The momentum density is evaluated at each point, so Π_a also depends on $\vec{x}$ and t . The reason we call this the momentum density is because here $\mathcal{L}$ is the Lagrangian density. In the same way, we can also define the Hamiltonian density associated with each degree of freedom (DOF):

$$\mathcal{H} = \Pi_a \dot{\phi}_a - \mathcal{L} \quad (8)$$

Then, the Hamiltonian you just integrate over all space

$$H = \int d^3x \mathcal{H} \quad (9)$$

essentially a sum over all degrees of freedom. So now we can talk about equation of motion. It is analogous to classical mechanics. The classical field theory is just like classical mechanics, you generalize it to infinite number of DOFs. So now with each point in space, you associate it with some DOFs. Again, you just do the variation of the action, you extremize the action to be 0,

$$\delta S = 0 \quad (10)$$

where the action is again defined by

$$S = \int dt L = \int d^4x \mathcal{L}(\phi_a, \partial_\mu \phi_a). \quad (11)$$

You can straightforwardly generalize this to include higher spatial derivatives. Now let us compute the variation:

$$0 = \delta S = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi_a} \delta \phi_a + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \delta (\partial_\mu \phi_a) \right] = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi_a} \delta \phi_a + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \partial_\mu (\delta \phi_a) \right] \quad (12)$$

In this case, δ is the variation, and ∂_μ and δ are independent, so you can exchange them, i.e.,

$$\partial_\mu (\delta \phi_a) = \delta (\partial_\mu \phi_a) \quad (13)$$

Then, we can do the integration by part for the second term and obtain

$$0 = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi_a} \delta \phi_a - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \right) \delta \phi_a \right] + \text{boundary terms} \quad (14)$$

The boundary terms will be proportional to $\delta \phi_a$. We will always choose $\delta \phi_a$ such that boundary terms vanish. Equation (14) must vanish for arbitrary variations $\delta \phi_a(x^\mu)$, which can be any function of x^μ . Therefore, the prefactor has to vanish, i.e.,

$$0 = \frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \right), \quad (15)$$

(15) is the general EOM for classical field theory.

Restrictions in Field Theory

From now on, we will make two restrictions for simplicity.

Translationally invariant

The first is that we restrict to field theory which are translationally invariant, which means the theory does not single out any special point in spacetime. If you perform the experiment here in Chengdu, it is the same as performing it in Washington DC. If the theory were not translationally invariant, experiments performed in Chengdu and Washington DC would have different results.

Lorentz invariant

The second is that we assume the field theory is Lorentz invariant. A Lorentz invariant quantity looks exactly the same for every observer moving at constant velocity, no matter how their coordinate axes are rotated or how fast they move relative to each other. A quantity is Lorentz invariant if it does not change under any Lorentz transformation that includes both: Spatial rotations (changing direction in space), and Lorentz boosts (changing velocity or reference frame).

Now let us look at some simple examples of classical field theories that satisfy these two conditions.

E.x. 1: Maxwell's theory.

The dynamical variable is the four-vector potential $A_\mu(x^\nu)$. For A_μ , you define the field strength

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad (16)$$

The electric and magnetic fields E and B can be obtained from $F_{\mu\nu}$. The action for E&M can be obtained by

$$S_{EM} = \int d^4x \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right) \quad (17)$$

This is the action for electromagnetism in the absence of charged matter. If you do the variation here to get the EOM, then you obtain the source-free (vacuum) Maxwell equations. This action is corresponding to a local theory due to a single integral of spacetime.

Lecture 3: Symmetry and Conservation Law

Ex 2: Einstein gravity Einstein gravity is another classical theory. The Einstein-Hilbert action is

$$S = \frac{1}{16\pi G_N} \int d^4x \sqrt{-g} R, \quad (18)$$

where g is the determinant of the metric and R is the Ricci scalar. Einstein gravity is a local theory involving a single spacetime integral over geometric quantities.

Ex 3: The simplest quantum field theory the scalar field theory: A_μ is the vector field in E&M, and in Einstein gravity the dynamical variable is the metric tensor $g_{\mu\nu}$, which is even more complicated. Thus, the simplest case to study is a scalar field.

Let us just consider the real-valued scalar field $\phi(x)$. At each spacetime point, the field takes a single value; thus $\phi(x)$ is the dynamical variable. In physics, there are many examples of scalar fields; e.g., the Higgs field or the pion field, whose excitations correspond to scalar particles.

The simplest family of scalar field actions has the following form:

$$S = \int d^4x \left[-\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi) \right], \quad (19)$$

which is compatible with locality, translational symmetry, and Lorentz symmetry. Here, you can see that the Lorentz indices are contracted. The Lorentz indices in the derivative are contracted, so this guarantees the theory Lorentz invariant. This is essentially the simplest quantum theory one can write down based on locality.

In general, we can take $V(\phi)$ to be a polynomial

$$V(\phi) = \text{function of } \phi \quad (20)$$

When a system has translational symmetry, it means that if you perform an experiment anywhere, the result will be the same. This implies that all parameters in $V(\phi)$ must be constant and cannot depend on spacetime. The canonical momentum is

$$\Pi(x) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \dot{\phi} \quad (21)$$

where $\dot{\phi}^2 = \partial_t^2 \phi$ due to

$$\partial_\mu \phi \partial^\mu \phi = -\dot{\phi}^2 + (\nabla \phi)^2 \quad (22)$$

And then the Hamiltonian density is

$$\mathcal{H} = \Pi \dot{\phi} - \mathcal{L} = \frac{\Pi^2}{2} + \frac{1}{2} (\nabla \phi)^2 + V(\phi), \quad (23)$$

which is expressed in terms of momentum. And EOM is given by

$$\partial^2 \phi - \frac{\partial V}{\partial \phi} = 0. \quad (24)$$

and again I'm using a shorthand notation,

$$\partial^2 = \partial_\mu \partial^\mu = -\partial_t^2 + \nabla^2. \quad (25)$$

Here, let me give some simple examples of the V . In the simplest case, take $V = f\phi$. From translational invariance, f must be constant and cannot depend on spacetime. Then, the EOM is

$$\partial^2 \phi = f, \quad (26)$$

In this case, essentially you have some kind of constant, which is source for this ϕ . In this case, f acts as a constant source for the field ϕ . We say the system has an external force f ; because if $f \neq 0$, then ϕ cannot vanish. In this sense, the field ϕ is always generated by f . However, we usually consider systems without external forces, where the field evolves on its own.

In that case, if we ignore external forces, the simplest situation is

$$V(\phi) = \frac{1}{2} m^2 \phi^2 \quad (27)$$

m has to be constant. In this case, you have the EOM:

$$\partial^2 \phi - m^2 \phi = 0, \quad (28)$$

This is the famous equation called the Klein-Gordon equation in classical field theory. And you can also consider more complicated field theories with $V(\phi)$ to be some higher polynomial or more complicated functions. But keep in mind, this square is special as then the EOM has linear term. However, when $V(\phi)$ includes terms beyond quadratic order, the equation becomes nonlinear, and the analysis becomes more complicated. Anyway, this would be the simplest theory we consider.

Symmetry

A **symmetry** is a transformation of the dynamical fields that leaves the action invariant. Formally, each field ϕ_a transforms into a function of all the fields:

$$\phi_a \longrightarrow \phi'_a(\phi_b), \quad (29)$$

and if the action S remains unchanged under this transformation, we say the theory has a symmetry.

Example: Translation symmetry. Consider the scalar field theory introduced in (19). If we perform a constant shift of spacetime coordinates,

$$x^\mu \longrightarrow x'^\mu = x^\mu + a^\mu, \quad (30)$$

where a^μ is a constant four-vector, the scalar field transforms as

$$\phi(x) = \phi'(x'), \quad (31)$$

meaning that the field value at a given spacetime point does not change, only its coordinate label does. You can verify that the scalar-field action is invariant under this translation, confirming that translation is indeed a symmetry.

Example: Lorentz symmetry. Similarly, a Lorentz transformation acts on spacetime as

$$x^\mu \longrightarrow x'^\mu = \Lambda^\mu{}_\nu x^\nu, \quad (32)$$

where $\Lambda^\mu{}_\nu$ is a constant Lorentz transformation matrix satisfying $\Lambda^T \eta \Lambda = \eta$. Under this transformation, the scalar field again obeys (31). The action remains invariant, so Lorentz invariance is another continuous spacetime symmetry.

Continuous and discrete symmetries. Transformations such as translations and Lorentz transformations depend on continuous parameters (a^μ and $\Lambda^\mu{}_\nu$), so they are called **continuous symmetries**. In four dimensions, a^μ has four continuous parameters, while $\Lambda^\mu{}_\nu$ contains six (three rotations and three boosts).

The same scalar-field action also admits **discrete symmetries**, for example:

$$\phi \longrightarrow -\phi, \quad (33)$$

$$\vec{x} \longrightarrow -\vec{x}, \quad (34)$$

corresponding to field inversion and spatial parity. Both leave the action invariant but involve no continuous parameters. Because performing either transformation twice returns the system to its original state, each corresponds to a $\mathbb{Z}_2$ symmetry.

Global and local symmetries. Symmetries are further classified as **global** or **local**:

- *Global symmetries* have spacetime-independent transformation parameters. Translations and Lorentz transformations are examples, since a^μ and $\Lambda^\mu{}_\nu$ are constants.
- *Local symmetries* have spacetime-dependent parameters. A well-known example is the gauge symmetry in electromagnetism, where the phase transformation depends on spacetime position.

Noethers theorem. A deep connection exists between continuous symmetries and conservation laws: each continuous global symmetry corresponds to a conserved quantity. In classical mechanics, time translation implies energy conservation, spatial translation implies momentum conservation, and rotational symmetry implies angular momentum conservation. This general relationship is formalized by **Noethers theorem**.

Consider a continuous symmetry with an infinitesimal parameter ϵ :

$$\phi_a \longrightarrow \phi'_a = \phi_a + \epsilon f_a(\phi_b, \partial_\mu \phi_b), \quad (35)$$

where f_a is a function of the fields and possibly their derivatives. The variation of the Lagrangian density under this infinitesimal transformation must be a total derivative:

$$\delta \mathcal{L} = \epsilon \partial_\mu K^\mu, \quad (36)$$

for some current K^μ . If $K^\mu = 0$, the Lagrangian is strictly invariant.

Using the Euler-Lagrange equations,

$$\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \right) = 0,$$

the variation of $\mathcal{L}$ can be written as

$$\hat{\delta} \mathcal{L} = \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \hat{\delta} \phi_a \right) = \epsilon \partial_\mu K^\mu. \quad (37)$$

Comparing both sides yields the conservation equation

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} f_a - K^\mu \right) = 0. \quad (38)$$

Defining the **Noether current**

$$J^\mu = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} f_a - K^\mu, \quad (39)$$

we obtain the conservation law

$$\partial_\mu J^\mu = 0. \quad (40)$$

Continuity equation and conserved charge. Writing $J^\mu = (J^0, \vec{J})$, the conservation law takes the familiar form

$$\partial_0 J^0 + \nabla \cdot \vec{J} = 0, \quad (41)$$

which expresses the local conservation of a density and its associated current. Integrating over all space gives

$$Q = \int d^3x J^0, \quad (42)$$

and differentiating (41) shows

$$\partial_0 Q = 0, \quad (43)$$

so the total charge Q is conserved. Equation (43) is the integrated form of the local continuity equation (41).

Remarks. This result is known as **Noethers first theorem**, valid when ϵ is spacetime independent (global symmetry). When the transformation parameter depends on spacetime, as in gauge theories, additional terms appear and the statement becomes **Noethers second theorem**, which relates local symmetries to redundancies among the equations of motion.

The key idea is that any finite continuous symmetry can be constructed by successively applying infinitesimal transformations. Hence, understanding the infinitesimal form of a symmetry is sufficient to determine its full finite structure.

Lecture 4 Why QFT?

With these preparations, we are ready to begin our discussion of **Quantum Mechanics**.

From Classical Mechanics to Quantum Mechanics

Recall that in **classical mechanics**, the simplest system consists of a single degree of freedom — a particle moving along one dimension, whose position is described by a function $x(t)$. The goal of classical mechanics is to *solve the equation of motion (EOM)* for $x(t)$. Once we find $x(t)$, we have completely described the dynamics of the system.

Quantization: From Variables to Operators

In **quantum mechanics**, the dynamical variable $x(t)$ becomes an **operator**

$$x(t) \longrightarrow \hat{x}(t),$$

and similarly, the conjugate momentum becomes $\hat{p}(t)$. The classical EOMs are replaced by **Heisenberg equations of motion** for these operators.

There are two equivalent but conceptually different pictures of quantum mechanics:

(a) Schrödinger Picture

- The **state** of the system is described by a wave function $\psi(x, t)$.
- Operators such as $\hat{x}$ and $\hat{p}$ are *time-independent*.
- The quantity x in $\psi(x, t)$ is the eigenvalue of $\hat{x}$.
- The time evolution of the system is governed by the **Schrödinger equation**.
- Once we know $\psi(x, t)$, we can compute expectation values and transition amplitudes.

(b) Heisenberg Picture

- The **operators** $\hat{x}(t)$ and $\hat{p}(t)$ are time-dependent.
- The **state vector** $|\psi\rangle$ is time-independent.
- Time evolution is governed by the **Heisenberg equation**

$$\frac{d\hat{O}}{dt} = i[\hat{H}, \hat{O}],$$

where $\hat{H}$ is the Hamiltonian.

- Once we solve the operator equations, expectation values can be evaluated in any state.

From Quantum Mechanics to Quantum Field Theory

In **field theory**, we proceed in an analogous way. The classical dynamical variables are now fields $\phi(\vec{x}, t)$, and upon quantization they become operators $\hat{\phi}(\vec{x}, t)$.

The key conceptual difference is that in classical mechanics, x is a *dynamical variable*, whereas in field theory, $\vec{x}$ merely labels spatial points — the *field* ϕ is the true dynamical variable.

In the Schrödinger picture of field theory:

$$\Psi[\phi(\vec{x}), t]$$

is a *wave functional* — a function of all possible field configurations $\phi(\vec{x})$. Here, $\phi(\vec{x})$ denotes the eigenvalues of the operator $\hat{\phi}(\vec{x})$, which does not depend on time in this picture. The Schrödinger equation determines the time evolution of $\Psi[\phi(\vec{x}), t]$.

In the Heisenberg picture, we focus instead on the operator equations of motion:

$$\hat{\phi}(\vec{x}, t), \quad \hat{\pi}(\vec{x}, t),$$

which satisfy the Heisenberg equations — the quantum analogues of the classical field equations. The state $|\psi\rangle$ remains fixed, and all dynamics are carried by the operators.

Remarks: In Quantum Field Theory, the Heisenberg picture is almost always used, because it directly generalizes classical field equations and simplifies computations. We will therefore adopt the Heisenberg picture from now on.

Why Relativistic Quantum Mechanics Fails

Before introducing QFT, let us comment briefly on **relativistic quantum mechanics**. Naively, one might attempt to combine quantum mechanics and special relativity directly. However, this approach fails for fundamental reasons, leading us naturally to quantum field theory.

From Schrödinger to Klein–Gordon

In non-relativistic quantum mechanics, we start from the dispersion relation

$$E = \frac{\vec{p}^2}{2m}. \quad (44)$$

Promoting energy and momentum to operators,

$$E \rightarrow i\frac{\partial}{\partial t}, \quad \vec{p} \rightarrow -i\nabla,$$

we obtain the Schrödinger equation

$$i\frac{\partial}{\partial t}\psi(\vec{x}, t) = -\frac{1}{2m}\nabla^2\psi(\vec{x}, t). \quad (45)$$

To include relativity, one might instead start with the relativistic dispersion relation

$$E^2 = \vec{p}^2 + m^2. \quad (46)$$

Applying the same operator substitutions yields

$$\partial_\mu\partial^\mu\psi(\vec{x}, t) - m^2\psi(\vec{x}, t) = 0, \quad (47)$$

the **Klein–Gordon equation**. Historically, this was regarded as the wave equation for a single relativistic particle.

Problems with the Klein–Gordon Interpretation

Soon, physicists realized several problems with interpreting (47) as a one-particle wave equation:

1. **No positive-definite probability density.** The conserved current derived from (47) cannot be interpreted as a non-negative probability density. Probability must be both non-negative and conserved — here, it fails the first condition.
2. **Negative-energy solutions.** From the dispersion relation

$$E = \pm\sqrt{\vec{p}^2 + m^2},$$

we see that quantum mechanics cannot simply discard the negative branch. If such states exist, the system becomes unstable: particles would cascade to states of ever-lower (negative) energy.

Deeper Conceptual Problem

Even if these issues were fixed, the entire concept of a wave function for a *fixed number of particles* breaks down in relativity. In non-relativistic systems, particle number is conserved, and a two-particle wave function $\psi(\vec{x}_1, \vec{x}_2, t)$ makes sense. But in relativistic systems, energy can create or annihilate particles: electrons and positrons can annihilate into photons, for example. Thus, the particle number is not fixed, and no single wave function can describe such a system.

The Necessity of Quantum Field Theory

To describe systems where particles can be created and destroyed, we must have a framework that can handle an arbitrary and variable number of particles in a unified way. This is precisely what **quantum field theory** provides.

In QFT, the fundamental objects are fields. Their excitations correspond to particles, automatically allowing for processes of creation and annihilation. Hence, QFT is not merely a generalization of quantum mechanics; it is the only consistent framework that unifies *quantum mechanics* with *special relativity*.

Field Theory from the Continuum Limit of Discrete Systems

The third motivation for Quantum Field Theory comes from the observation that **field theories can also emerge as the continuum limit of discrete systems**. This perspective is especially relevant in **condensed matter physics**.

Example: A Chain of Coupled Oscillators

Consider a one-dimensional chain of particles (for instance, atoms in a crystal lattice) connected by springs. Let the equilibrium spacing between neighboring atoms be a , and label the positions of the atoms by $x_0, x_1, x_2, \dots$. The displacement of the n -th atom from its equilibrium position is denoted by η_n :

$$x_n - x_n^{(0)} = \eta_n. \quad (48)$$

The Lagrangian of this discrete system is

$$L = T - V = \sum_n \left(\frac{1}{2} \mu \dot{\eta}_n^2 - \frac{\lambda}{2} (\eta_{n+1} - \eta_n)^2 - \frac{1}{2} \sigma \eta_n^2 \right), \quad (49)$$

where

- T is the kinetic energy and V is the potential,
- μ is the mass of each particle,
- λ characterizes the spring coupling between nearest neighbors,
- σ describes the restoring potential trapping each atom near equilibrium.

This is a simple yet instructive model of coupled harmonic oscillators.

Continuum Limit

As the lattice spacing $a \rightarrow 0$, the chain becomes a continuous medium. We then replace the discrete displacements $\eta_n(t)$ by a smooth function $\eta(x, t)$, where $x = na$ labels the position along the chain.

In this limit, the discrete sum becomes an integral:

$$a \sum_n \rightarrow \int dx. \quad (50)$$

The Lagrangian (49) becomes

$$\begin{aligned} L &= \sum_n a \left(\frac{\mu}{2a} \dot{\eta}_n^2 - \frac{a\lambda}{2} \left(\frac{\eta_{n+1} - \eta_n}{a} \right)^2 - \frac{\sigma}{2a} \eta_n^2 \right) \\ &= \int dx \left(\frac{\tilde{\mu}}{2} \dot{\eta}^2 - \frac{\tilde{\lambda}}{2} (\partial_x \eta)^2 - \frac{\tilde{\sigma}}{2} \eta^2 \right), \end{aligned} \quad (51)$$

where we define

$$\tilde{\mu} = \frac{\mu}{a}, \quad \tilde{\lambda} = \frac{\lambda}{a}, \quad \tilde{\sigma} = \frac{\sigma}{a}. \quad (52)$$

To obtain a well-defined continuum theory, we keep $\tilde{\mu}$, $\tilde{\lambda}$, and $\tilde{\sigma}$ fixed as $a \rightarrow 0$.

The Emergent Field Theory

The continuum Lagrangian can be rewritten as

$$L = \tilde{\mu} \int dx \left(\frac{1}{2} \dot{\eta}^2 - \frac{\tilde{\lambda}}{2\tilde{\mu}} (\partial_x \eta)^2 - \frac{\tilde{\sigma}}{2\tilde{\mu}} \eta^2 \right) = \tilde{\mu} \int dx \left(\frac{1}{2} \dot{\eta}^2 - \frac{v^2}{2} (\partial_x \eta)^2 - \frac{m^2}{2} \eta^2 \right), \quad (53)$$

where we define

$$v^2 = \frac{\tilde{\lambda}}{\tilde{\mu}}, \quad m^2 = \frac{\tilde{\sigma}}{\tilde{\mu}}. \quad (54)$$

This Lagrangian is precisely the form of a **scalar field theory** discussed earlier. If we set $v = 1$, it becomes relativistic and corresponds to the case where the propagation speed equals the speed of light. In general, for $v \neq 1$, it describes a **non-relativistic field theory**.

Physical Interpretation

Even though this model is extremely simple, it captures a fundamental idea: many condensed matter systems — such as vibrations in a solid — can be described by lattice models at the microscopic level, but by continuum field theories at macroscopic scales.

When we further include quantum mechanics, the quantization of such continuum fields naturally leads to **quantum field theory**. Thus, QFT is not only a framework for particle physics but also the natural language for describing collective excitations in many-body systems.

Summary: All Paths Lead to QFT

- Quantum Field Theory describes the **quantum dynamics of classical fields** (e.g., electromagnetic, gravitational, scalar fields).
- It **unifies special relativity and quantum mechanics** in a consistent way.
- It also **emerges as the continuum limit of discrete systems**, relevant to many areas such as condensed matter physics.

In short, whether we begin from quantum mechanics, relativity, or discrete many-body systems, **all roads lead to Quantum Field Theory**.

Lecture 5. Quantization of the Harmonic Oscillator and Free Scalar Field

In this lecture, we will discuss how to quantize a theory, starting from the familiar example of the harmonic oscillator and then generalizing to field theory. The key idea is that a free scalar field is essentially an infinite collection of harmonic oscillators labeled by the wave vector $\vec{k}$.

Quantization of the Harmonic Oscillator in the Heisenberg Picture

The Lagrangian for a one-dimensional harmonic oscillator (with mass $m = 1$) is

$$L = \frac{1}{2}\dot{x}^2 - \frac{1}{2}\omega^2 x^2. \quad (55)$$

The conjugate momentum is $p = \dot{x}$, and the Hamiltonian becomes

$$H = p\dot{x} - L = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 x^2. \quad (56)$$

The equation of motion is

$$\ddot{x} + \omega^2 x = 0. \quad (57)$$

The general classical solution is

$$x(t) = A \cos(\omega t) + B \sin(\omega t), \quad (58)$$

where A and B are integration constants. Equivalently, we can write it in complex form:

$$x(t) = \frac{1}{\sqrt{2}} (a e^{-i\omega t} + a^* e^{i\omega t}), \quad (59)$$

where a is a complex integration constant.

Quantization: To quantize the system, we promote the classical variables to quantum operators in the Heisenberg picture:

$$x(t) \rightarrow \hat{x}(t), \quad p(t) \rightarrow \hat{p}(t). \quad (60)$$

The operator equation of motion is identical in form to the classical one,

$$\ddot{\hat{x}} + \omega^2 \hat{x} = 0, \quad (61)$$

but its interpretation changes since $\hat{x}$ is now an operator. The general operator solution mirrors the classical one:

$$\hat{x}(t) = \hat{A} \cos(\omega t) + \hat{B} \sin(\omega t) = \frac{1}{\sqrt{2}} (\hat{a} e^{-i\omega t} + \hat{a}^\dagger e^{i\omega t}), \quad (62)$$

where $\hat{a}$ and $\hat{a}^\dagger$ are constant operators (time-independent in the Heisenberg picture). The momentum operator follows from differentiation:

$$\hat{p}(t) = \dot{\hat{x}}(t) = \frac{-i\omega}{\sqrt{2}} (\hat{a} e^{-i\omega t} - \hat{a}^\dagger e^{i\omega t}). \quad (63)$$

Canonical Quantization: We impose the canonical commutation relation

$$[\hat{x}(t), \hat{x}'(t)] = [\hat{p}(t), \hat{p}'(t)] = 0, \quad [\hat{x}(t), \hat{p}(t)] = i, \quad (64)$$

where we assume $\hbar = 1$. This implies the fundamental algebra for the creation and annihilation operators:

$$[\hat{a}, \hat{a}] = [\hat{a}^\dagger, \hat{a}^\dagger] = 0, \quad [\hat{a}, \hat{a}^\dagger] = 1 \quad (65)$$

Hilbert Space Construction: The vacuum state is defined as

$$\hat{a}|0\rangle = 0, \quad (66)$$

and the excited states are obtained by applying $\hat{a}^\dagger$ repeatedly:

$$|n\rangle = \frac{1}{\sqrt{n!}} (\hat{a}^\dagger)^n |0\rangle. \quad (67)$$

These states form a complete orthonormal basis for the Hilbert space of the harmonic oscillator.

Summary of the Quantization Procedure: The general steps for quantizing a system are:

1. Start with the classical equation of motion.
2. Find the most general classical solution.
3. Promote classical dynamical variables to quantum operators.
4. Impose canonical commutation relations.
5. Use the resulting constant operators to construct the Hilbert space.

This procedure applies equally well to systems with many degrees of freedom, including field theories.

Quantization of the Free Scalar Field

We now apply the same logic to a relativistic field with infinitely many degrees of freedom. Consider the free scalar field with action

$$S = \int d^4x \left[\frac{1}{2} (\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2} m^2 \phi^2 \right]. \quad (68)$$

The equation of motion is

$$(\partial^2 + m^2)\phi = 0. \quad (69)$$

The canonical momentum and Hamiltonian density are

$$\Pi = \dot{\phi}, \quad \mathcal{H} = \frac{1}{2} \Pi^2 + \frac{1}{2} (\nabla \phi)^2 + \frac{1}{2} m^2 \phi^2. \quad (70)$$

Classical Solutions: Because of translation symmetry, we can expand ϕ in Fourier modes:

$$\phi(x) = e^{-iEt + i\vec{k} \cdot \vec{x}}. \quad (71)$$

Substituting into (69) gives the dispersion relation

$$E^2 = \vec{k}^2 + m^2, \quad (72)$$

Here we define the frequency $\omega_{\vec{k}}$

$$\omega_{\vec{k}} \equiv \sqrt{\vec{k}^2 + m^2}. \quad (73)$$

Then, we define the positive-frequency and negative-frequency solutions:

$$u_{\vec{k}}(x) = e^{-i\omega_{\vec{k}}t + i\vec{k} \cdot \vec{x}}, \quad (74)$$

$$u_{\vec{k}}^*(x) = e^{i\omega_{\vec{k}}t - i\vec{k} \cdot \vec{x}}. \quad (75)$$

The most general classical solution is

$$\phi(x) = \int \frac{d^3\vec{k}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \left(a_{\vec{k}} u_{\vec{k}}(x) + a_{\vec{k}}^* u_{\vec{k}}^*(x) \right). \quad (76)$$

Quantization: Promote ϕ and Π to operators:

$$\hat{\phi}(x) = \int \frac{d^3\vec{k}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \left(\hat{a}_{\vec{k}} u_{\vec{k}}(x) + \hat{a}_{\vec{k}}^\dagger u_{\vec{k}}^*(x) \right). \quad (77)$$

The canonical commutation relations generalize to

$$[\hat{\phi}(t, \vec{x}), \hat{\phi}(t, \vec{x}')] = 0, \quad (78)$$

$$[\hat{\Pi}(t, \vec{x}), \hat{\Pi}(t, \vec{x}')] = 0, \quad (79)$$

$$[\hat{\phi}(t, \vec{x}), \hat{\Pi}(t, \vec{x}')] = i\delta^{(3)}(\vec{x} - \vec{x}'). \quad (80)$$

From these, one can derive

$$[\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}'}] = 0, \quad (81)$$

$$[\hat{a}_{\vec{k}}^\dagger, \hat{a}_{\vec{k}'}^\dagger] = 0, \quad (82)$$

$$[\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}'}^\dagger] = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}'). \quad (83)$$

Interpretation: Each mode $\vec{k}$ behaves as an independent quantum harmonic oscillator with frequency $\omega_{\vec{k}}$. Therefore, a free scalar field can be viewed as an infinite set of harmonic oscillators, one for each $\vec{k}$.

Hilbert Space: The vacuum state $|0\rangle$ satisfies

$$\hat{a}_{\vec{k}}|0\rangle = 0, \quad \forall \vec{k}. \quad (84)$$

Excited states are created by acting with $\hat{a}_{\vec{k}}^\dagger$:

$$|\vec{k}_1, \vec{k}_2, \dots, \vec{k}_n\rangle = \hat{a}_{\vec{k}_1}^\dagger \hat{a}_{\vec{k}_2}^\dagger \cdots \hat{a}_{\vec{k}_n}^\dagger |0\rangle. \quad (85)$$

These states represent n -particle states of the quantized field.

Conclusion: Quantizing the free scalar field follows the exact same steps as quantizing a harmonic oscillator. Each mode of the field behaves as an independent harmonic oscillator with its own frequency $\omega_{\vec{k}}$. The resulting quantum field is therefore an infinite collection of oscillators, and its excitations correspond to particles.

Lecture 6: Hamiltonian and Zero Point Energy in Free Scalar Field Theory

Last time, we quantized the free scalar field theory with the action

$$S = \int d^4x \left(-\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 \right). \quad (86)$$

The conjugate momentum density and the Hamiltonian density are

$$\Pi = \partial_t \phi, \quad \mathcal{H} = \frac{1}{2} \Pi^2 + \frac{1}{2} (\nabla \phi)^2 + \frac{1}{2} m^2 \phi^2. \quad (87)$$

The classical equation of motion is

$$(\partial^2 - m^2) \phi = 0. \quad (88)$$

Following the general quantization procedure, we promote ϕ and Π to operators and obtain the complete solution to the operator equation:

$$\hat{\phi}(x) = \int \frac{d^3 \vec{k}}{(2\pi)^3} \frac{1}{\sqrt{\omega_{\vec{k}}}} \left[\hat{a}_{\vec{k}} u_{\vec{k}} + \hat{a}_{\vec{k}}^\dagger u_{\vec{k}}^* \right], \quad (89)$$

where $u_{\vec{k}}$ is a complete set of mode functions:

$$u_{\vec{k}}(x) = e^{-i\omega_{\vec{k}}t + i\vec{k} \cdot \vec{x}}, \quad \omega_{\vec{k}} = \sqrt{\vec{k}^2 + m^2}. \quad (90)$$

The conjugate momentum density is

$$\hat{\Pi}(x) = \partial_t \hat{\phi}(x) = -i \int \frac{d^3 \vec{k}}{(2\pi)^3} \sqrt{\omega_{\vec{k}}} \left[\hat{a}_{\vec{k}} u_{\vec{k}} - \hat{a}_{\vec{k}}^\dagger u_{\vec{k}}^* \right]. \quad (91)$$

Canonical Commutation Relations: The fundamental commutation relations are

$$[\hat{\phi}(t, \vec{x}), \hat{\phi}(t, \vec{x}')] = [\hat{\Pi}(t, \vec{x}), \hat{\Pi}(t, \vec{x}')] = 0, \quad [\hat{\phi}(t, \vec{x}), \hat{\Pi}(t, \vec{x}')] = i\delta^{(3)}(\vec{x} - \vec{x}'). \quad (92)$$

From these, one finds

$$[\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}'}] = [\hat{a}_{\vec{k}}^\dagger, \hat{a}_{\vec{k}'}^\dagger] = 0, \quad [\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}'}^\dagger] = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}'). \quad (93)$$

Each mode $\vec{k}$ behaves as an independent harmonic oscillator. The commutator (93) is a continuum generalization of the familiar $[\hat{a}, \hat{a}^\dagger] = 1$ relation for a single oscillator. Different $\vec{k}$ -modes commute because of the delta function.

Hamiltonian in Terms of $\hat{a}_{\vec{k}}$ and $\hat{a}_{\vec{k}}^\dagger$

The total Hamiltonian is obtained by integrating the Hamiltonian density:

$$H = \int d^3x \mathcal{H} = \int d^3x \left[\frac{1}{2} \hat{\Pi}^2 + \frac{1}{2} (\nabla \hat{\phi})^2 + \frac{1}{2} m^2 \hat{\phi}^2 \right]. \quad (94)$$

Substituting the field expansions, one finds

$$H = \frac{1}{2} \int \frac{d^3 \vec{k}}{(2\pi)^3} \omega_{\vec{k}} \left(\hat{a}_{\vec{k}}^\dagger \hat{a}_{\vec{k}} + \hat{a}_{\vec{k}} \hat{a}_{\vec{k}}^\dagger \right) = \int \frac{d^3 \vec{k}}{(2\pi)^3} \omega_{\vec{k}} \left(\hat{a}_{\vec{k}}^\dagger \hat{a}_{\vec{k}} + \frac{1}{2} [\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}}^\dagger] \right). \quad (95)$$

Using the commutation relation (93), we obtain

$$H = \int \frac{d^3 \vec{k}}{(2\pi)^3} \omega_{\vec{k}} \hat{a}_{\vec{k}}^\dagger \hat{a}_{\vec{k}} + E_0, \quad (96)$$

where the vacuum energy (zero-point energy) is

$$E_0 = \int \frac{d^3 \vec{k}}{(2\pi)^3} \frac{1}{2} \omega_{\vec{k}} (2\pi)^3 \delta^{(3)}(\vec{k}) (0). \quad (97)$$

Interpretation of $\delta_{(\vec{k})}^{(3)}(0)$: Recall the definition of the delta function:

$$(2\pi)^3 \delta^{(3)}(\vec{k}) = \int d^3x e^{i\vec{k}\cdot\vec{x}}. \quad (98)$$

Setting $\vec{k} = 0$, we find

$$(2\pi)^3 \delta^{(3)}(0) = \int d^3x = V, \quad (99)$$

which represents the spatial volume of the system. Hence, the zero-point energy can be written as

$$E_0 = V \epsilon_0, \quad \epsilon_0 = \int \frac{d^3\vec{k}}{(2\pi)^3} \frac{1}{2} \omega_{\vec{k}}, \quad (100)$$

where ϵ_0 is the vacuum energy density.

Divergence of ϵ_0 : Substituting $\omega_{\vec{k}} = \sqrt{\vec{k}^2 + m^2}$, we have

$$\epsilon_0 = \frac{1}{2} \int \frac{d^3\vec{k}}{(2\pi)^3} \sqrt{\vec{k}^2 + m^2}, \quad (101)$$

which is divergent.

Physically, this divergence arises because quantum field theory has a *continuum* of degrees of freedom. If we imagine a discrete lattice model with spacing a , each lattice point hosts a harmonic oscillator. In the continuum limit $a \rightarrow 0$, any finite volume contains infinitely many oscillators, leading to divergent total zero-point energy.

This is the first example of a divergence in QFT. Dealing with such divergences and understanding how they appear and how to remove or renormalize them is one of the major challenges and defining features of quantum field theory.

Physical Interpretation and Conserved Quantities

Time Independence of H : Even though $\hat{\phi}$ and $\hat{\Pi}$ are time-dependent operators, the Hamiltonian H itself is conserved because of time translation symmetry. The time integration is not performed in constructing H , only a spatial integration is done. The conservation of H follows from Noethers theorem applied to time translation symmetry.

Momentum Operator: Spatial translation symmetry gives rise to another conserved Noether charge:

$$P^i = - \int d^3x \Pi(t, \vec{x}) \partial_i \phi(t, \vec{x}). \quad (102)$$

In terms of the creation and annihilation operators, this becomes

$$P^i = \int \frac{d^3\vec{k}}{(2\pi)^3} k^i \hat{a}_{\vec{k}}^\dagger \hat{a}_{\vec{k}}. \quad (103)$$

Unlike the Hamiltonian, there is no zero-point contribution in P^i , since the vacuum state carries zero momentum.

The theory also possesses Lorentz invariance, which leads to conserved angular momentum and boost charges $M^{\mu\nu}$. These too can be expressed in terms of $\hat{a}_{\vec{k}}$ and $\hat{a}_{\vec{k}}^\dagger$.

Regularization and the Cosmological Constant Problem

To make sense of the divergent zero-point energy, we introduce a momentum cutoff Λ :

$$\epsilon_0 = \frac{1}{2} \int^\Lambda \frac{d^3\vec{k}}{(2\pi)^3} \sqrt{\vec{k}^2 + m^2}, \quad (104)$$

which restricts the integral to momenta $|\vec{k}| < \Lambda$. This cutoff corresponds to the scale at which our effective field theory ceases to be valid. In a lattice model, $\Lambda \sim 1/a$.

Even with the cutoff, ϵ_0 remains a very large number. Physically, this zero-point energy is measurable in principle, but we do not observe such an enormous vacuum energy in the universe. This discrepancy between theoretical prediction and observation is known as the **cosmological constant problem**.

Remarks

- The Hamiltonian is time-independent due to time translation symmetry.
- The quantum field theory of ϕ reduces to a continuum of independent harmonic oscillators labeled by $\vec{k}$, each with frequency $\omega_{\vec{k}} = \sqrt{m^2 + \vec{k}^2}$.
- This picture connects naturally with a lattice of coupled oscillators: the continuum limit ($a \rightarrow 0$) yields infinitely many oscillators, one per point in space.
- The divergences encountered here are a direct consequence of the continuum nature of fields. Handling these divergences is central to modern quantum field theory.

Lecture 7. Hilbert Space and Particle Interpretation in Free Scalar QFT

In the free scalar field theory, each momentum mode behaves like an independent harmonic oscillator. This allows us to construct the Hilbert space of the theory in direct analogy with the harmonic oscillator in quantum mechanics.

Vacuum State and Fock-Space Construction

For each momentum mode $\vec{k}$, we introduce the operators $\hat{a}_{\vec{k}}$ and $\hat{a}_{\vec{k}}^\dagger$. The **vacuum state** is defined by

$$\hat{a}_{\vec{k}} |0\rangle = 0, \quad \forall \vec{k}. \quad (105)$$

We postulate the existence of such a state. Later, one can explicitly write the wave functional of $|0\rangle$ in terms of $\phi(\vec{x})$, similar to the ground-state wave function of the harmonic oscillator.

Once the vacuum is defined, all states are obtained by acting with creation operators:

$$|n_{\vec{k}_1}, n_{\vec{k}_2}, \dots\rangle \propto \left(\hat{a}_{\vec{k}_1}^\dagger\right)^{n_{\vec{k}_1}} \left(\hat{a}_{\vec{k}_2}^\dagger\right)^{n_{\vec{k}_2}} \dots |0\rangle. \quad (106)$$

Thus the Hilbert space is the tensor product of infinitely many harmonic oscillators, one for each momentum $\vec{k}$.

Single- and Two-Particle States

The simplest excited state is

$$|\vec{k}\rangle = \hat{a}_{\vec{k}}^\dagger |0\rangle. \quad (107)$$

Two excitations give

$$|\vec{k}_1, \vec{k}_2\rangle = \hat{a}_{\vec{k}_1}^\dagger \hat{a}_{\vec{k}_2}^\dagger |0\rangle. \quad (108)$$

Because the creation operators commute,

$$[\hat{a}_{\vec{k}_1}^\dagger, \hat{a}_{\vec{k}_2}^\dagger] = 0,$$

all multi-particle states are symmetric under exchange. Therefore, the quanta of a real scalar field are **bosons**.

Energy–Momentum Eigenvalues

Ignoring the zero-point energy E_0 , the vacuum satisfies

$$H|0\rangle = 0, \quad P^i|0\rangle = 0. \quad (109)$$

One may show (exercise for students) that

$$H|\vec{k}\rangle = \omega_{\vec{k}}|\vec{k}\rangle, \quad P^i|\vec{k}\rangle = k^i|\vec{k}\rangle, \quad (110)$$

where $\omega_{\vec{k}} = \sqrt{\vec{k}^2 + m^2}$. Thus the one-particle state carries the 4-momentum

$$p^\mu = (\omega_{\vec{k}}, \vec{k}),$$

which satisfies $p^2 = -m^2$. Hence the excitation created by $\hat{a}_{\vec{k}}^\dagger$ is interpreted as a relativistic particle of mass m .

For the two-particle state,

$$H|\vec{k}_1, \vec{k}_2\rangle = (\omega_{\vec{k}_1} + \omega_{\vec{k}_2})|\vec{k}_1, \vec{k}_2\rangle, \quad (111)$$

$$\vec{P}|\vec{k}_1, \vec{k}_2\rangle = (\vec{k}_1 + \vec{k}_2)|\vec{k}_1, \vec{k}_2\rangle. \quad (112)$$

The energies simply add, reflecting the fact that the free theory has no interactions.

Normalization and Lorentz Covariance

The plane-wave states satisfy

$$\langle \vec{k} | \vec{k}' \rangle = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}'). \quad (113)$$

This normalization is convenient but not Lorentz covariant. A better choice is the covariantly normalized state

$$|k\rangle = \sqrt{2\omega_{\vec{k}}} |\vec{k}\rangle, \quad (114)$$

for which

$$\langle k | k' \rangle = 2\omega_{\vec{k}} (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}'). \quad (115)$$

The factor $\omega_{\vec{k}} \delta^{(3)}(\vec{k} - \vec{k}')$ transforms simply under Lorentz transformations.

If U_{Λ} denotes the unitary operator associated with a Lorentz transformation Λ , then

$$U_{\Lambda} |k\rangle = |\Lambda k\rangle. \quad (116)$$

Physical Wave Packets

The states $|\vec{k}\rangle$ and $|k\rangle$ are not normalizable, just like plane waves $e^{i\vec{k}\cdot\vec{x}}$ in non-relativistic quantum mechanics. A **physical** one-particle state is a wave packet,

$$|\Psi\rangle = \int \frac{d^3\vec{k}}{(2\pi)^3} f(\vec{k}) |k\rangle, \quad (117)$$

with normalization

$$\langle \Psi | \Psi \rangle = \int \frac{d^3\vec{k}}{(2\pi)^3} |f(\vec{k})|^2 < \infty. \quad (118)$$

Choosing $f(\vec{k})$ appropriately, one can construct localized wave packets. Similarly, a two-particle wave packet takes the form

$$\int \frac{d^3\vec{k}_1}{(2\pi)^3} \frac{d^3\vec{k}_2}{(2\pi)^3} f(\vec{k}_1, \vec{k}_2) |k_1, k_2\rangle. \quad (119)$$

Structure of the Hilbert Space: Fock Space

Because the theory is free, the Hilbert space factorizes as

$$\mathcal{H} = \mathcal{H}_0 \oplus \mathcal{H}_1 \oplus \mathcal{H}_2 \oplus \cdots, \quad (120)$$

where $\mathcal{H}_0$ is the vacuum sector, $\mathcal{H}_1$ the one-particle sector, $\mathcal{H}_2$ the two-particle sector, and so on. The direct sum of all sectors with finite particle number is the **Fock space**.

Remarks

- Each momentum mode is a harmonic oscillator. Each excitation corresponds to a particle.
- Since $[\hat{a}_{\vec{k}_1}^\dagger, \hat{a}_{\vec{k}_2}^\dagger] = 0$, all particles are bosons.
- All physical energies are positive, despite $E^2 = \vec{k}^2 + m^2$ admitting $\pm\omega_{\vec{k}}$.
- In the free theory, the total energy is the sum of the individual energies; there is no interaction energy.
- Plane-wave states are not normalizable, so physical states must be constructed as wave packets.
- Although $\vec{k}$ labels a continuum, the space of normalizable one-particle states is separable and admits a countable orthonormal basis, just as in ordinary quantum mechanics.

Lecture 8 Conserved Charges, Noether's Theorem, and Complex Scalar Fields

In the previous lectures we quantized the free scalar field and interpreted its excitations as relativistic particles. We also discussed the structure of the Hilbert space and the role of conserved quantities such as energy and momentum. In this lecture we study Noether's theorem in field theory, the quantum meaning of conserved charges, and the appearance of particles and anti-particles in the complex scalar field.

Noether's Theorem and Conserved Currents

Consider a set of fields ϕ_a and a continuous symmetry acting on them:

$$\phi_a \longrightarrow \phi'_a = \phi_a + \epsilon_\alpha f_a^\alpha(\phi, \partial\phi). \quad (121)$$

Here:

- ϵ_α are infinitesimal parameters, one for each symmetry,
- a labels the fields, and α labels the symmetries.

If the action is invariant under this transformation up to a total derivative,

$$\delta\mathcal{L} = \epsilon_\alpha \partial_\mu K^\mu_\alpha, \quad (122)$$

then Noether's theorem tells us that the current

$$J^\mu_\alpha = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)} f_a^\alpha - K^\mu_\alpha \quad (123)$$

is conserved:

$$\partial_\mu J^\mu_\alpha = 0. \quad (124)$$

The associated conserved charge is

$$Q_\alpha = \int d^3x J^0_\alpha. \quad (125)$$

The index position on α is irrelevant; it simply labels the symmetry.

Quantum Interpretation: Charges Generate Symmetries

At the quantum level the fields become operators $\hat{\phi}_a$, the current becomes $\hat{J}^\mu_\alpha$, and the conserved charge becomes an operator

$$\hat{Q}_\alpha = \int d^3x \hat{J}^0_\alpha.$$

Since $\partial_\mu J^\mu_\alpha = 0$, the quantum charge is time independent:

$$\frac{d\hat{Q}_\alpha}{dt} = 0.$$

The fundamental result is that the operator $\hat{Q}_\alpha$ generates the symmetry. For infinitesimal transformations,

$$[\epsilon_\alpha \hat{Q}_\alpha, \hat{\phi}_a(x)] = -i\epsilon_\alpha f_a^\alpha(x). \quad (126)$$

Simplified Proof for Internal Symmetries

For internal symmetries, f_a^α typically contains no time derivatives and $K^\mu_\alpha = 0$. In this case,

$$\hat{J}^0_\alpha = \hat{\Pi}_a f_a^\alpha,$$

where

$$\Pi_a = \frac{\partial\mathcal{L}}{\partial(\partial_0\phi_a)} \longrightarrow \text{operator } \hat{\Pi}_a$$

after quantization. Using the equal-time commutation relation

$$[\hat{\phi}_a(t, \vec{x}), \hat{\Pi}_b(t, \vec{x}')] = i\delta_{ab} \delta^{(3)}(\vec{x} - \vec{x}'), \quad (127)$$

one obtains

$$[\epsilon_\alpha \hat{Q}_\alpha, \hat{\phi}_a(t, \vec{x})] = -i\epsilon_\alpha f_a^\alpha.$$

Example: Time Translations A useful example is the symmetry under time translations. Recall that the Hamiltonian density is defined as

$$\hat{\mathcal{H}} = \hat{\Pi}_a \partial_t \hat{\phi}_a - \mathcal{L}. \quad (128)$$

For a time translation, the change of the Lagrangian is a total derivative, so $\mathcal{L}$ plays the role of the quantity K^μ_α in $\delta\mathcal{L} = \epsilon_\alpha \partial_\mu K^\mu_\alpha$. The corresponding conserved charge is the Hamiltonian operator

$$\hat{H} = \int d^3x \hat{\mathcal{H}}. \quad (129)$$

Using the equal-time commutation relations, we find

$$[\hat{H}, \hat{\phi}_a(t, \vec{x})] = -i \partial_t \hat{\phi}_a(t, \vec{x}), \quad (130)$$

which is exactly the Heisenberg equation of motion. This shows that $\hat{H}$ acts as the generator of time translations. Infinitesimally, a time translation changes the field by

$$\delta \hat{\phi}_a = \epsilon \partial_t \hat{\phi}_a,$$

and this transformation is precisely generated by the commutator with $\hat{H}$.

Finite symmetry transformation

A finite symmetry transformation is generated by the unitary operator

$$\hat{U}(\Lambda_\alpha) = \exp\left(i\Lambda_\alpha \hat{Q}_\alpha\right), \quad (131)$$

which acts on the field operator as

$$\hat{U}(\Lambda_\alpha) \hat{\phi}_a(x) \hat{U}^\dagger(\Lambda_\alpha) = \hat{\phi}'_a(x). \quad (132)$$

Expanding $\hat{U}$ to first order in Λ_α reproduces the infinitesimal result (126).

Example: Time Translations The finite time translation is produced by the unitary operator

$$\hat{U}(t) = e^{i\hat{H}t}, \quad (133)$$

and we have

$$\hat{U}(t) \hat{\phi}(0, \vec{x}) \hat{U}^\dagger(t) = \hat{\phi}(t, \vec{x}), \quad (134)$$

which is the operator statement that $\hat{H}$ generates time evolution.

Complex Scalar Field and Anti-Particles

Consider the complex scalar Lagrangian

$$\mathcal{L} = -\partial_\mu \phi \partial^\mu \phi^\dagger - m^2 \phi^\dagger \phi. \quad (135)$$

The momenta are

$$\hat{\Pi}_\phi = \partial_0 \hat{\phi}^\dagger, \quad \hat{\Pi}_{\phi^\dagger} = \partial_0 \hat{\phi}.$$

The mode expansion of the quantum field is

$$\hat{\phi}(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \left[\hat{a}_{\vec{k}} e^{-i\omega_k t + i\vec{k} \cdot \vec{x}} + \hat{b}_{\vec{k}}^\dagger e^{i\omega_k t - i\vec{k} \cdot \vec{x}} \right], \quad (136)$$

with

$$[\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}'}^\dagger] = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}'), \quad [\hat{b}_{\vec{k}}, \hat{b}_{\vec{k}'}^\dagger] = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}'). \quad (137)$$

The vacuum satisfies

$$\hat{a}_{\vec{k}}|0\rangle = \hat{b}_{\vec{k}}|0\rangle = 0.$$

Thus there are two sets of quanta:

$$\hat{a}_{\vec{k}}^\dagger|0\rangle, \quad \hat{b}_{\vec{k}}^\dagger|0\rangle,$$

both with mass m .

The U(1) Charge and Particle, Anti-Particle Interpretation

The Lagrangian is invariant under the phase rotation

$$\phi \rightarrow e^{i\alpha}\phi.$$

The Noether charge operator is

$$\hat{Q} = i \int d^3x (\hat{\Pi}_{\phi^\dagger} \hat{\phi}^\dagger - \hat{\Pi}_\phi \hat{\phi}). \quad (138)$$

or

$$\hat{Q} = i \int d^3x (\hat{\phi}^\dagger \hat{\Pi}_{\phi^\dagger} - \hat{\phi} \hat{\Pi}_\phi). \quad (139)$$

which has the ordering ambiguity. To resolve ordering ambiguities, we impose

$$\hat{Q}|0\rangle = 0.$$

One then obtains

$$\hat{Q} = \int \frac{d^3k}{(2\pi)^3} (\hat{a}_{\vec{k}}^\dagger \hat{a}_{\vec{k}} - \hat{b}_{\vec{k}}^\dagger \hat{b}_{\vec{k}}). \quad (140)$$

Thus:

$$[\hat{Q}, \hat{a}_{\vec{k}}^\dagger] = +\hat{a}_{\vec{k}}^\dagger, \quad (141)$$

$$[\hat{Q}, \hat{b}_{\vec{k}}^\dagger] = -\hat{b}_{\vec{k}}^\dagger. \quad (142)$$

Define covariantly normalized states:

$$|k\rangle = \sqrt{2\omega_k} \hat{a}_{\vec{k}}^\dagger |0\rangle, \quad |\bar{k}\rangle = \sqrt{2\omega_k} \hat{b}_{\vec{k}}^\dagger |0\rangle.$$

Then

$$\hat{Q}|k\rangle = +|k\rangle, \quad \hat{Q}|\bar{k}\rangle = -|\bar{k}\rangle.$$

Thus:

- The $\hat{a}^\dagger$ quanta carry charge +1,
- The $\hat{b}^\dagger$ quanta carry charge -1,
- They have the same mass and spin zero.

These are interpreted as a **particle** and its **anti-particle**.

Remark. In relativistic QFT every particle has an associated anti-particle. For a real scalar field, $\hat{\phi}^\dagger = \hat{\phi}$, so the particle is its own anti-particle.